CFD Analysis

Flow Analysis in FlightStream Software

FlightStream: Surface Vorticity Solver

- Based on potential flow theory and vorticity transport equation

Governing Equations

- **Potential Flow Equation:** Assumes irrotational flow
- **Vorticity Transport Equation:** Describes evolution of vorticity in the flow field
- **Boundary Conditions:** Ensure flow tangency condition on the surface of the body
- **Panel Method**
  - Discretize surface into small panels
  - Calculate induced flow by summing contributions from all panels

Potential Flow Equation

$$\nabla^2 \phi = 0$$

Vorticity Transport Equation:

$$\frac{\partial \omega}{\partial t} + (\mathbf{u} \cdot \nabla)\omega = (\omega \cdot \nabla)\mathbf{u} + \nu \nabla^2 \omega$$

Boundary Conditions:

$$\mathbf{u} \cdot \mathbf{n} = 0$$

Panel Method:

$$\mathbf{u}(\mathbf{r}) = \frac{1}{4\pi} \int_S \frac{\gamma \times (\mathbf{r} - \mathbf{r_t})}{|\mathbf{r} - \mathbf{r_t}|^3} \, dS_t$$

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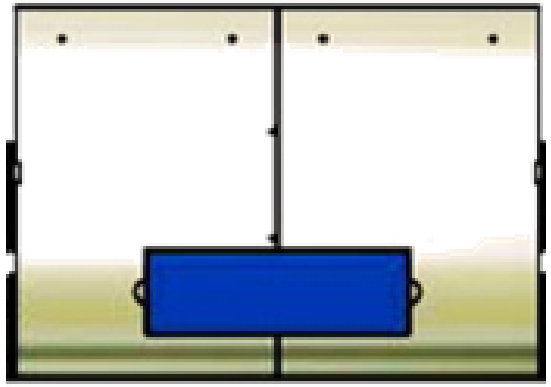
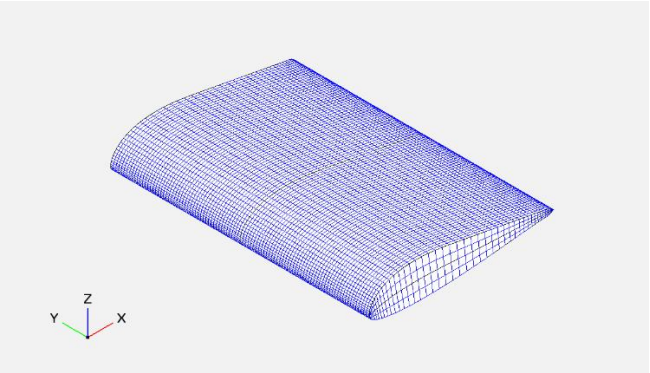
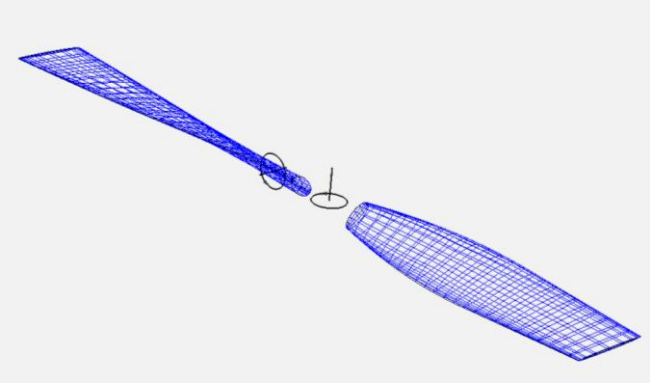
Geometry and mesh generation for wing and propeller

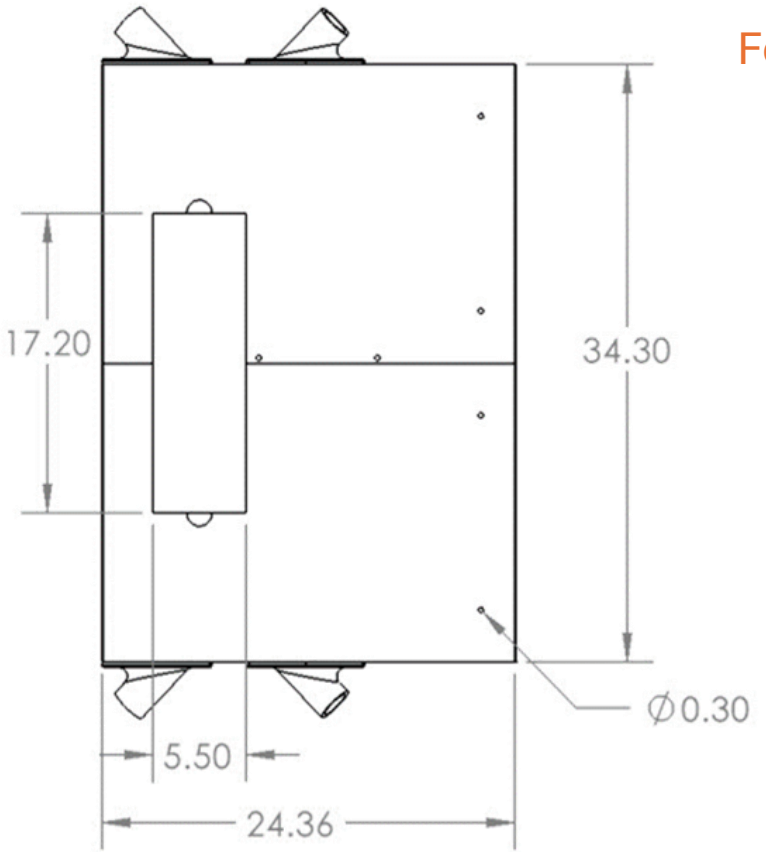
- Primary mesh elements: triangles and quadrilaterals.
- Rectangle mesh elements used for wing and propeller geometry.
- Tessellation values in OpenVSP:
  - U: divisions along the longitudinal direction.
  - W: divisions along the circumferential direction.



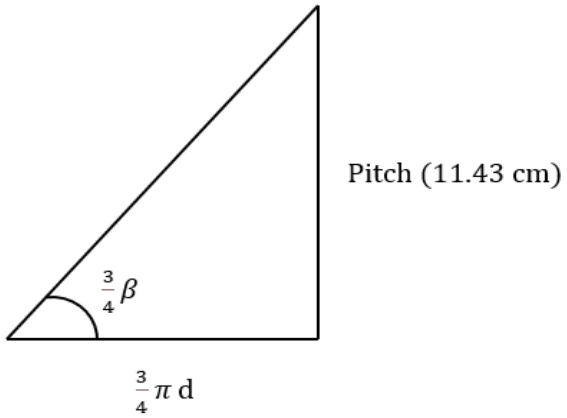




Sizing of Wing and Propeller



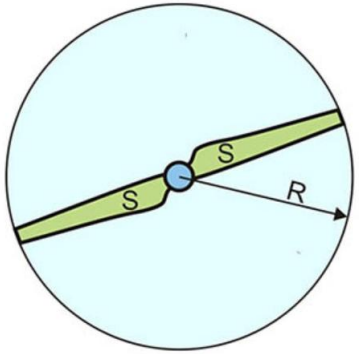
Pitch:
Forward distance traveled by a propeller in one complete revolution



$$\frac{3}{4}\beta = tan^{-1}\left(\frac{11.43}{\frac{3}{4}\pi d}\right)$$

$$\beta = 12^{\circ}$$

Solidity Ratio:
Proportion of total disk area occupied by the blades



2 blade wind rotor

$$\sigma = \frac{Nc}{\pi R} = 0.1257$$

β – Pitch,
σ – Solidity Ratio,
R- Radius,
N- No of Rotor Blades,
c – Mean Aerodynamic Chord

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Mesh Quality Assessment and Refinement

Mesh Element Size Selection
- 5 x 5 grid chosen
- 25 simulations conducted

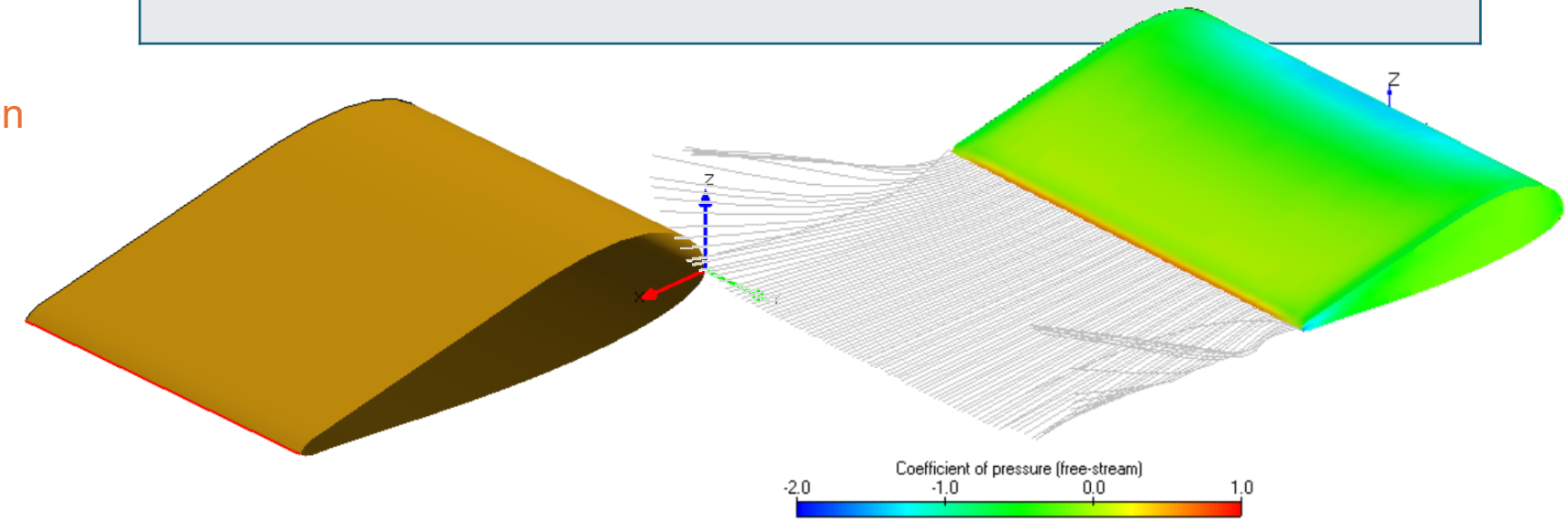
Spanwise Division Values
- Varied from 10 to 50

Chordwise Division Values
- Varied from 13 to 53

Boundary Conditions

- Freestream Boundary Condition
- No-Slip Boundary Condition
- Symmetry Boundary Condition
- Wall Boundary Condition

Parameter	Value
Angle of attack (deg)	2 to 16
Freestream Velocity (m/s)	0.1
Number of Iterations	500
Convergence Threshold	1.00e-05
Reference Velocity (m/s)	0.1
Reference Length (m)	0.25
Reference Area (m^2)	0.0875
Minimum coefficient of pressure	-100
Wake termination location (m)	0.5



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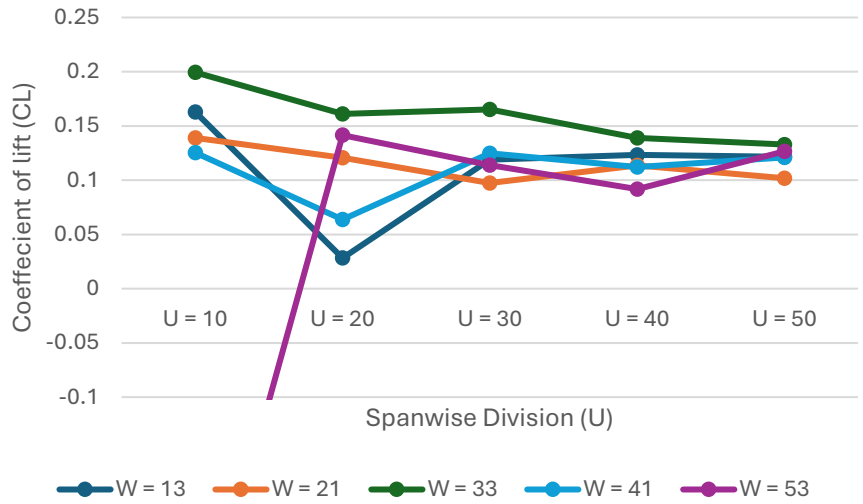
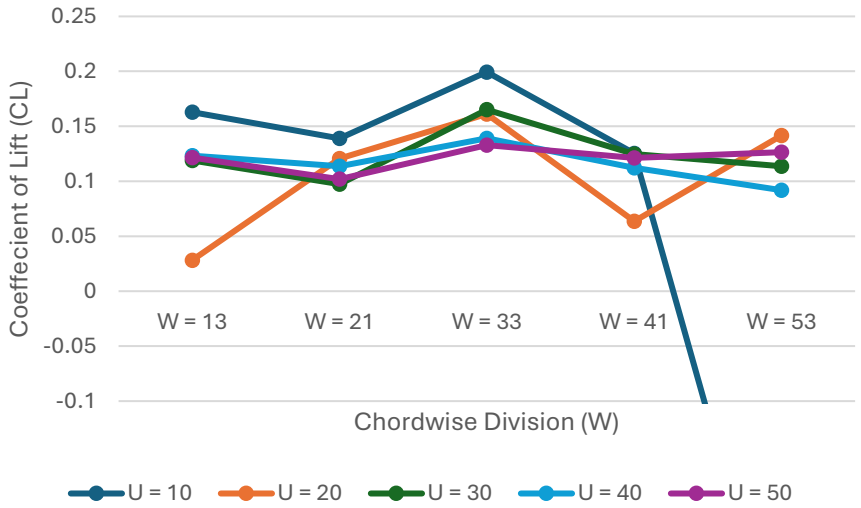
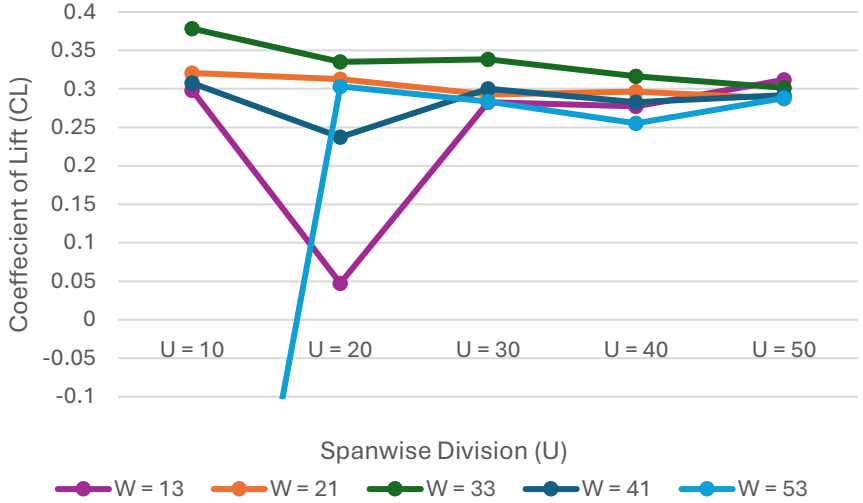
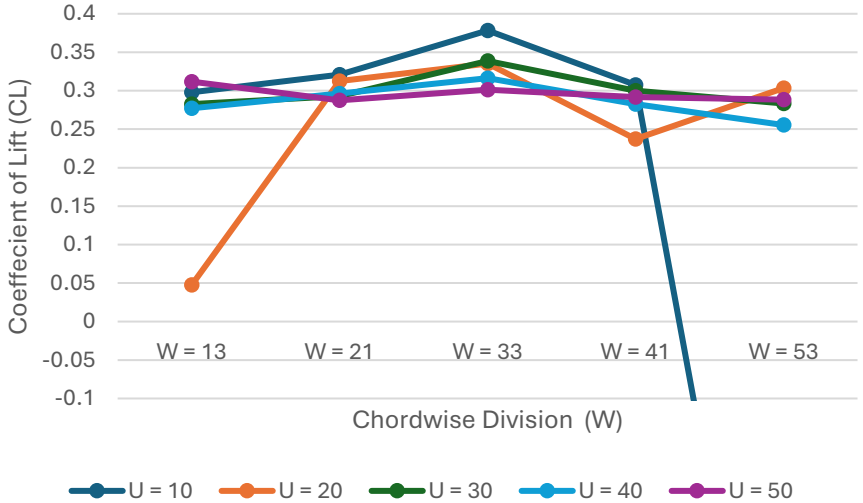
Mesh Quality Assessment and Refinement





Cl values for 2-degree AOA at different mesh resolutions





Cl values for 8-degree AOA at different mesh resolutions

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Force and Coefficient Data for Selected Tesellation

Angle of Attack (AOA)	Force coefficient along x-axis (Cx)	Force coefficient along y-axis (Cy)	Force coefficient along x-axis (Cz)	Coefficient of lift (CL)	Coefficient of induced drag (Cdi))	Zero lift drag coefficient (Cdo))
2	0.1098	0	0.1772	0.1733	0.011	0.1049
4	0.1083	0	0.2406	0.2325	0.0186	0.1063
6	0.1038	0	0.2931	0.2807	0.026	0.1079
8	0.1121	0	0.3652	0.346	0.0387	0.1231
10	0.1012	0	0.4145	0.3906	0.0503	0.1213
12	0.1017	0	0.4539	0.4229	0.0623	0.1315
14	0.0987	0	0.5093	0.4703	0.0774	0.1416
16	0.0918	0	0.5777	0.5301	0.0985	0.149

Aerodynamic coefficients for a wing section having U = 50, W = 41 mesh tessellation

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Validation of CFD Data with Static Thrust Stand Test

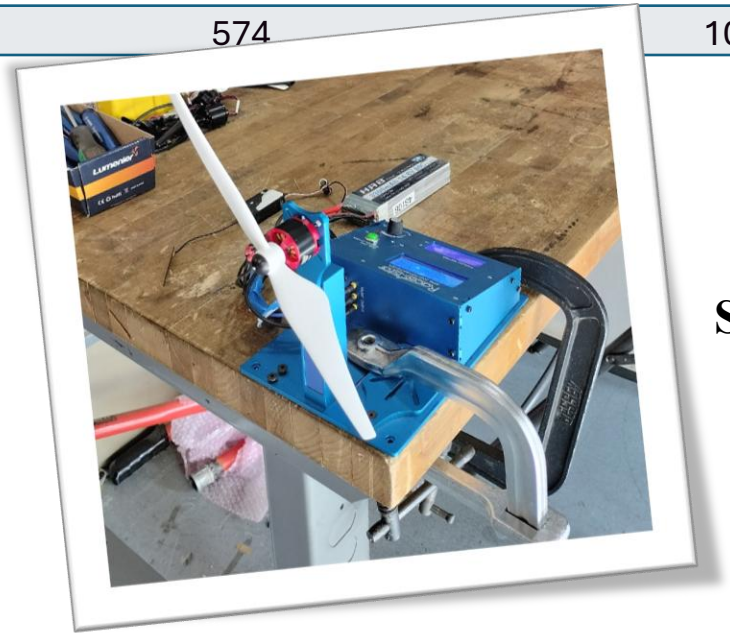
Validation Process

- Conducted static propeller thrust test in the lab.
- Used the same propulsion unit as in-flight testing.

Data Utilization

- RPM and power data obtained.
- Used in rotor simulations study in FlightStream software.

RPM	Thrust generated (gf)	Single motor power (W)	Overall Power (W)
5000	175	24	96
5400	209	29	116
5800	240	36	144
6200	280	42.3	169.2
6600	318	50.7	202.8
7000	347	54.1	216.4
7400	382	63.2	252.8
7800	433	75.3	301.2
8200	492	87	348
8600	515	94	376
9000	574	109.3	437.2



Static thrust test stand set up

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Comparison of Tested Static Thrust Values to Computed Values

Parameter	Value
Time Increment (sec)	0.0003 (Varying with RPM)
Time Stepping Iteration	72
Angular Velocity (rad/s)	586.43 (Varying with RPM)
Angle of attack (deg)	0
Freestream Velocity (m/s)	0.1
Number of Iterations	500
Convergence Threshold	1.00e-05
Reference Velocity (m/s)	0.1
Minimum coefficient of pressure	-100

RPM	Thrust from test stand (gf)	Angular Velocity (rad/sec)	Thrust from CFD analysis (N)	Thrust from CFD analysis (gf)	% Difference between test stand & CFD Analysis
5600	220	586.5066	2.5521	260.1529	18.2513
6000	250	628.4	2.887	294.2915	17.7166
6400	290	670.2933	3.3224	338.6748	16.7844
6800	327	712.1866	3.7877	386.1060	18.0752
7200	377	754.08	4.2662	434.8827	15.3535
7600	408	795.9733	4.7497	484.1692	18.6689
8000	449	837.8666	5.304	540.6727	20.4171

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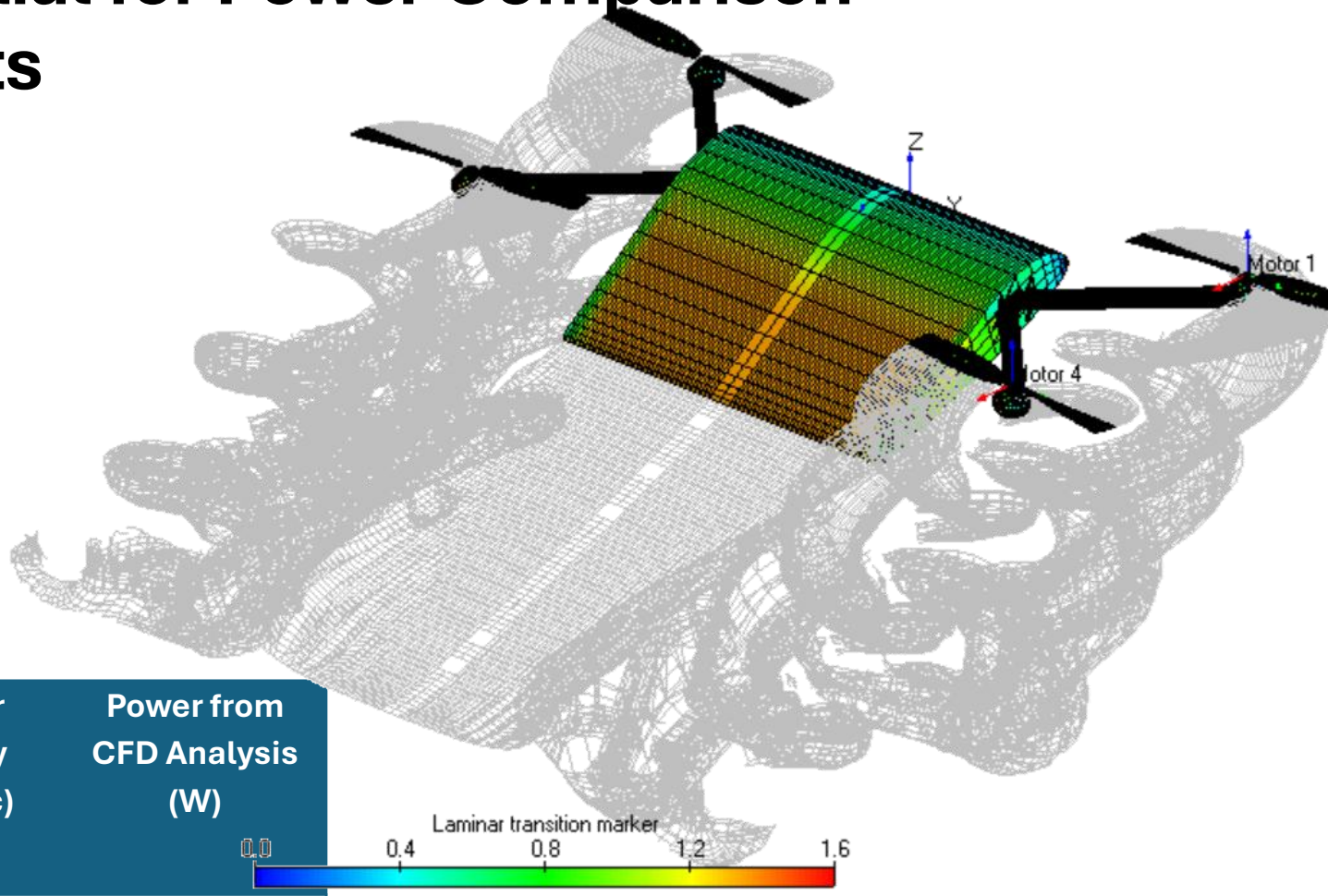
Full Body Simulation: Essential for Power Comparison between CFD and Flight Tests

Simulation Purpose

- Compare power between CFD and flight tests

Assumption

- Angular velocity for motors obtained from static thrust test data
- Ideally should be obtained from experimental flight test logs



Velocity (m/s)	Experimental Power Consumption (W)	Overall Torque of Motors from CFD Analysis (Nm)	Angular Velocity (rad/sec)	Power from CFD Analysis (W)
7.5	146	0.6724	607.37	408.39
10	124	0.2156	586.43	**126.43**
15	138	0.633	596.9	377.8

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Wind Tunnel Testing of Quadfoil in Low-Speed Tunnel Section at ODU

Objectives :-

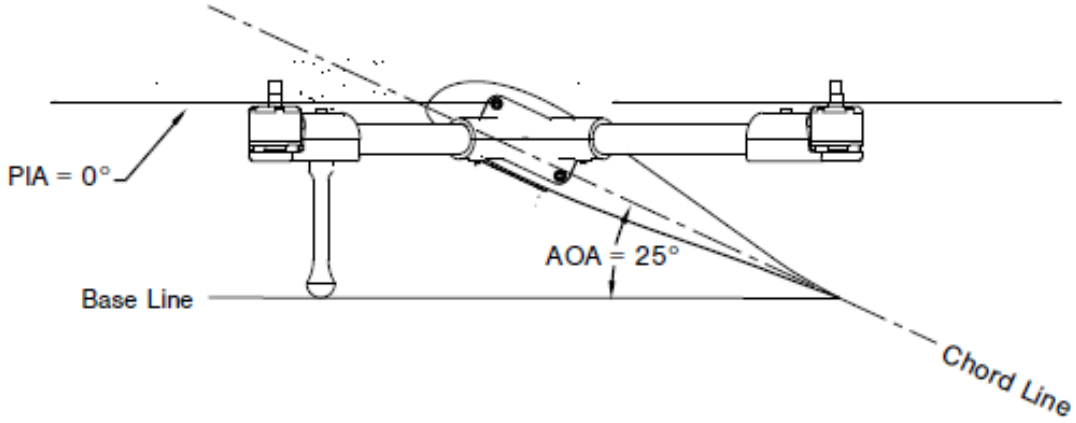
1. To find forces acting on the drone about 3 conventional axis with varying velocity.
2. To measure power required to rotate the motors at varying velocity.
3. To obtain trim condition parameters at varying velocity.
4. To evaluate the additional efficiency provided by incorporation of airfoil in the center body.
5. To assess the maneuvering characteristics of the design at 10 m/s velocity.

Note : - Tests were conducted after my graduation.

Equipment's Used and Test Carried Out

Equipment Used:-

1. Aircraft – Quadfoil
2. Connector – 25 deg fixed AOA arm connector
3. Propellers – a. 9.4 x 5 propeller , b. 10 x 4.5 propeller

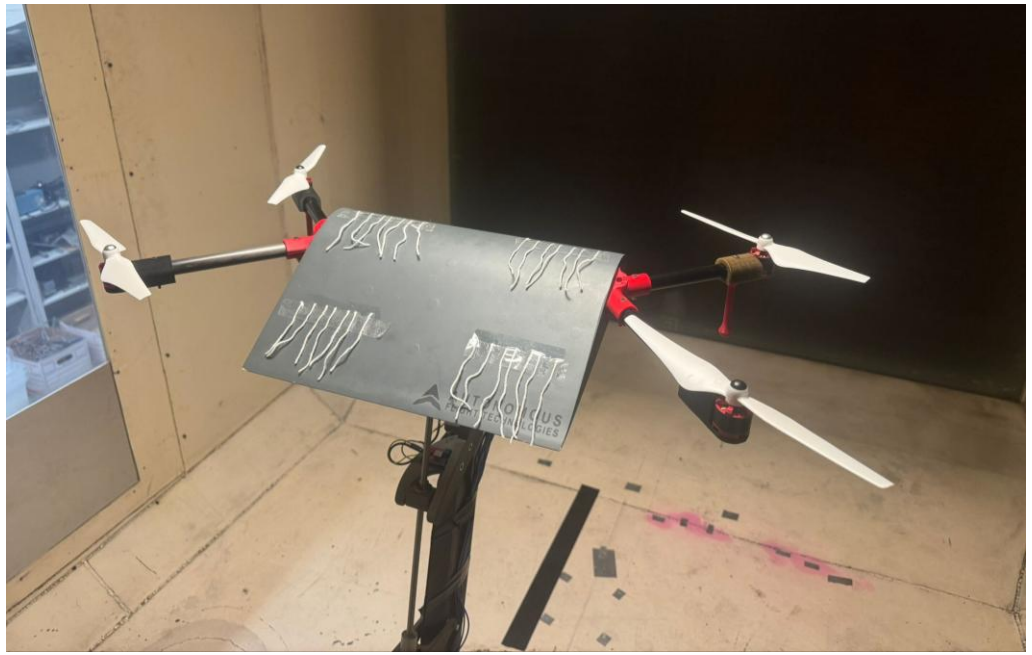


Reference figure for PIA

Tests Carried Out :-

1. Dynamic rotor test with 9.4 x 4.5 propeller at velocity 0, 2.5, 5, 7.5, 10, 12 m/s.
2. Dynamic rotor test with 10 x 4.5 propeller at velocity 0, 2.5, 5, 7.5, 10, 12 m/s.
3. Static rotor test with props off the body at velocity 0, 2.5, 5, 7.5, 10, 12 m/s.
4. Tuft flow visualization test.
5. Dynamic rotor test with pitch sweep (herein Propeller Incidence Angle (PIA) was varied from -15 deg to +10 deg)
6. Roughness test to force the transition in the flow (laminar to turbulent). It may help in delaying flow separation.

Images for Clarity



Tuft flow test

Roughness test

Note : - The tufts were cut in half to prevent them from getting tangled (for the actual test).

Design of Experiment (DOE) to Find Trim Condition Values

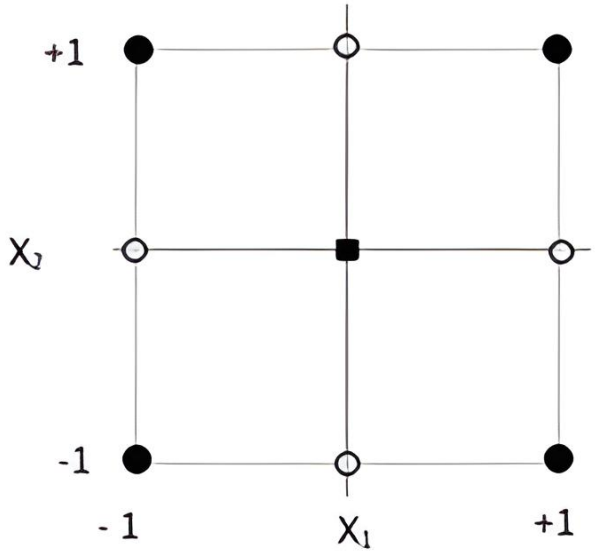
- Wind tunnel test matrix was designed to bracket the trim conditions measured in flight testing.
- α_p, n_f and n_r were varied over a regular grid of points known as the face-centered central composite design (FCD).

The central composite design is arguably the most popular second order design from response surface methodology (RSM).

The regression model equation can be written as

$$y = \beta 0 + \beta 1x1 + \beta 2x2 + \beta 12x1x2 + \beta 11x_1^2 + \beta 22x_2^2 + \epsilon$$

Where, y - Response (dependent variable)
x1 and x2 - Independent variables.
β's - Regression coefficients
ϵ - Random error.



Face centered central composite design in two factors

- The trim conditions for the level flight are

1. L = W = 2.86 lb (or 1.3 kg)
2. D = 0
3. $M_{CG} = 0$

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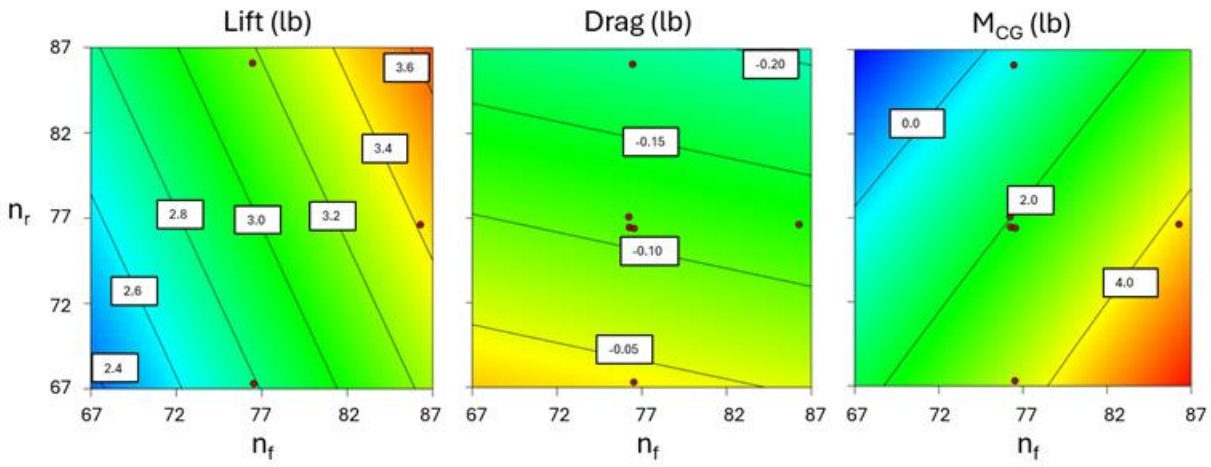
Aerospace Graduate Resea

Tunnel Test Matrix Used for V = 10 m/s & its Corresponding Response Contour

α_p	n_f	n_r	L (lbs)	D (lbs)	M (in-lb)	Power (W)
-15	76.1	76.3	3.11	-0.04	2.26	143
-15	66.7	86.4	2.94	-0.09	-1.39	147
-15	86.2	66.2	3.35	0.05	5.91	143
-15	66.5	66.5	2.49	0.05	1.77	97
-15	86.3	86.3	3.77	-0.08	2.70	192
-15	76.5	86.1	3.96	-0.19	1.81	163
-15	86.1	86.1	3.30	0.19	1.63	144
-17	76.5	76.4	2.98	-0.1	2.18	139
-17	76.5	76.5	2.78	-0.04	3.52	166
-17	67.3	67.3	3.01	0.19	1.98	147
-17	86.3	86.3	3.02	0.19	2.11	139
-17	76.6	76.6	2.96	-0.1	2.09	139
-17	76.6	76.6	2.55	-0.07	2.18	120
-17	86.3	86.3	2.94	-0.09	1.65	97
-17	86.0	86.0	3.00	-0.04	1.77	144
-19	66.5	66.5	2.22	-0.09	1.65	97
-19	86.7	86.7	3.47	-0.32	5.79	192
-19	66.7	76.2	2.59	-0.27	-1.62	147
-19	66.5	66.5	2.22	-0.09	1.65	97
-19	86.7	86.7	3.47	-0.32	5.79	192
-19	86.0	86.0	3.47	-0.32	5.79	192
-19	66.7	66.7	2.59	-0.27	-1.62	147
-19	86.6	86.6	2.75	-0.2	2.03	139
-19	76.2	76.2	2.75	-0.2	2.03	139

The pitch/PIA (α_p) data from the flight test and RPS (n_f and n_r) data from the static thrust test was used to design the 17 runs seen in the table on the left. The α_p varied from -15 to -19 deg, n_f and n_r varied from 67-87 RPS.



Response contours for v= 10 m/s using 9.4 x 5-inch propeller

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Results from V = 10 m/s Test Run

Regression coefficients at v = 10 m/s using 9.4 x 5-inch propeller

V = 10 (m/s)	L (lbs)	D (lbs)	M (in-lb)	Power (W)
Intercept	-0.61	0.32	2	118.84
αp	0.08	-0.02	0.56	21.65
n_f	0.04	-0.01	0.14	-1.88
n_r	0.02	-0.01	-0.05	2.46
$\alpha^2{}_p$	-	-	0.02	0.63
$n^2{}_f$	-	-	-	0.03

Summary of fit statistics for v = 10 m/s using 9.45-inch propeller

V = 10 (m/s)	L (lbs)	D (lbs)	M_{CG} (in-lb)	Power (W)
Standard Deviation	0.03	0.01	0.03	1.04
Mean	2.98	0.11	2.12	143.1
R^2	1	0.98	1	1
Adjusted R^2	1	0.98	1	1
Predicted R^2	1	0.97	1	1

Optimizer constraints and solution for v = 10 m/s using 9.45-inch propeller

Constraints				Solution
Name	**Goal**	**Lower Limit**	**Upper Limit**	**Trim Condition**
αp	in range	-19	-14	-14.01
n_f	in range	67	87	67.05
n_r	in range	67	87	79.43
Lift (lb)	target = 2.86			2.86
Drag (lb)	target = 0			0
M_{CG} **(in-lb)**	target = 0			0
Power (W)	none			133.75

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Trim Condition Data for Quadfoil UAV

Trim condition power data for 9.4 x 4.5 propeller from the wind tunnel study.

- **Table B1 data** is derived from nonlinear regression model executed in MATLAB.
- **Table B2 data** is derived from regression model created using Ordinary Least Square (OLS) method, executed in Design Expert software by Old Dominion University researchers. The power value was calculated from regression model equations for each of the velocities.

Velocity (m/s)	Trim PIA	Front rps	Rear rps	Lift (lb)	Drag (lb)	My (in-lb)	Trim Power (W)
0	-0.4	88.97	86.54	2.86	0	0	204.07
2.5	-5	84.4	87.4	2.86	0	0	186.79
5	-10	81.04	86.01	2.86	0	0	169.72
7.5	-12	77.46	85.21	2.86	0	0	161.91
10	-14	67.29	79.47	2.86	0	0	133.16
12	-17	66.9	79.12	2.86	0	0	125.7

Table B1

Velocity (m/s)	Trim PIA	Front rps	Rear rps	Lift (lb)	Drag (lb)	My (in-lb)	Trim Power (W)
0	-0.5	88.86	86.32	2.86	0	0	201.15
2.5	-5	83.97	86.95	2.86	0	0	184.01
5	-8.4	81.17	86.01	2.86	0	0	170.9
7.5	-12	77.42	85.16	2.86	0	0	160.94
10	-14	67.05	79.43	2.86	0	0	133.78
12	-17	66.71	78.7	2.86	0	0	124.21

Table B2

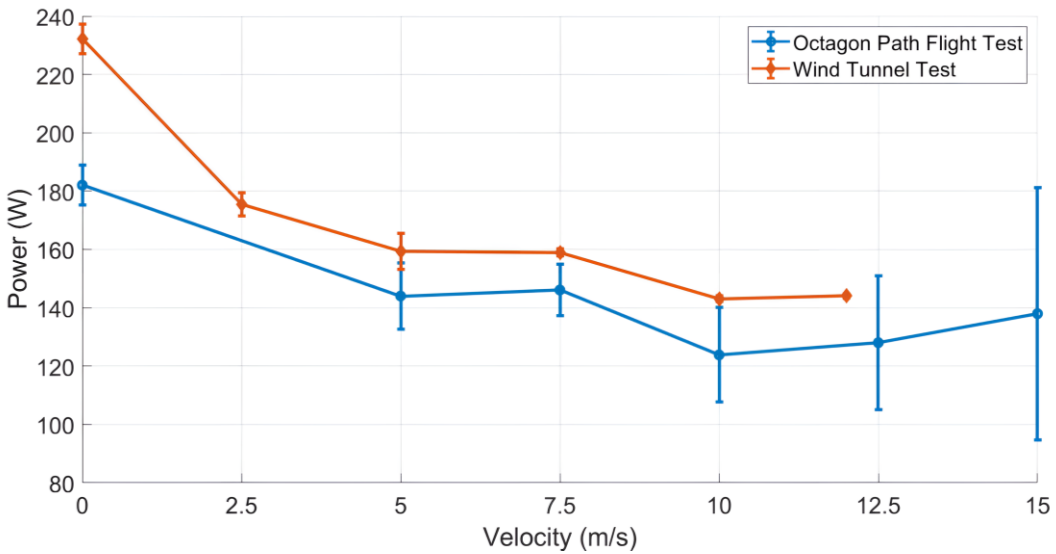
- Both models provide similar values. ODU's model (Table B2 data) can be considered since there is **enough literature** for it. This data is crucial for **CFD simulation model** and for **designing flight control laws** for this new configuration.

Power Comparison between Flight Test and Wind Tunnel Test

Power comparison between octagon path flight test data and tunnel test data at various velocities.

- Table B3 includes the power data along with **standard deviation** to assess the **uncertainty** at each velocity.

Velocity (m/s)	Octagon Path Flight Test		Wind Tunnel Test	
	Power (W)	Standard Deviation	Power (W)	Standard Deviation
0	182.04	6.77	232.12	5.1
2.5			175.42	3.95
5	144	11.3	159.38	6.18
7.5	146.07	8.75	158.98	1.28
10	123.83	16.2	143.1	1.04
12			122.05	0.78
12.5	128.02	23		
15	137.96	43.3		



Power comparison with uncertainty

The trend for power from both the test is similar. However, there is a discrepancy at v = 0m/s. The local density while conducting flight test was reported to be 1.107 kg/ m^3 , which is lower than the local density while conducting tunnel test (i.e. 1.21 kg/m^3). As per the power consumed during induced hover formula , the flight test power must be higher than the tunnel power. Probable cause for this discrepancy is recirculation effect.

$$P = \frac{T^{3/2}}{\sqrt{2\rho A}}$$

Where,
T – Thrust in N or lbf, ρ - Density in kg/m or slugs/ ft^3
P – Power in W or hp, A – Rotor disc area m^2 or $inch^2$

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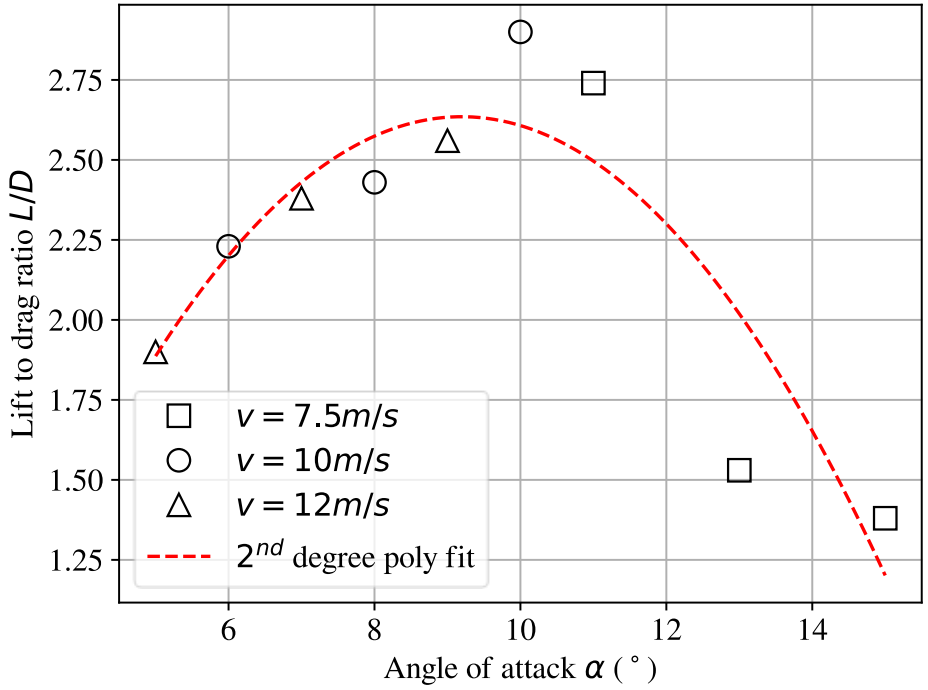
Prop Off Study to Understand the Aerodynamic Advantage of Wing

Velocity (m/s)	AOA (deg)	Lift (lbf)	Drag (lbf)	L/D ratio
7.5	15	0.21	0.16	1.31
7.5	13	0.24	0.15	1.6
7.5	11	0.33	0.12	2.75
10	10	0.55	0.19	2.89
10	8	0.46	0.19	2.42
10	6	0.38	0.17	2.24
12	5	0.44	0.23	1.91
12	7	0.61	0.26	2.35
12	9	0.71	0.28	2.54



This data can be compared to the CFD simulation results, the mesh resolution as well as boundary conditions of the simulations could be improved if there is larger percentage error.

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